PatientTriage.ai

BUSINESS PROPOSAL

Intelligent Emergency Care Orchestration

A clinical decision support and operational intelligence platform for the emergency department — combining deterministic clinical rules, AI-assisted assessment, and continuous deterioration monitoring, with the clinician always in control.

PREPARED FOR THE ACCENTURE INNOVATION CHALLENGE

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Status: Working prototype complete — see accompanying source repository and live demo

Classification: Illustrative business proposal. Market and industry figures are cited from public sources; in-product performance figures (agreement rates, throughput, ROI) are clearly labeled as illustrative simulations, not measured clinical outcomes.

Table of Contents

How to read this document. Section 2 establishes why emergency departments need better triage and monitoring tooling, citing independent published research. Section 3 describes what was actually built — a working, fully-tested prototype, not a mockup. Sections 4–8 make the business case for taking it further: who it serves, what it could be worth, how it would be rolled out safely, and what could go wrong along the way.

SECTION 1

Executive Summary

Emergency departments are the front door of the health system, and that front door is under sustained strain. PatientTriage.ai is a working prototype of a platform that helps emergency nursing and physician teams triage faster, more consistently, and with continuous visibility into patient deterioration — without ever taking the final clinical decision away from a human clinician.

The prototype was built for the Accenture Innovation Challenge to demonstrate a specific point of view: that the highest-value application of AI in emergency care is not to replace clinical judgment, but to make it faster to exercise, easier to explain, and safer to fall back from when the AI itself is wrong, unavailable, or simply less certain than the nurse standing at the bedside. Every AI-generated recommendation in the product sits alongside a deterministic, rule-based clinical engine (Emergency Severity Index logic, NEWS2, qSOFA) that keeps functioning even if the AI service is degraded or offline, and every recommendation requires an explicit human confirmation, override, or escalation before it affects patient flow.

The result is a five-workspace application — **Operations, Triage, Monitoring, Capacity,** and **Analytics** — that carries a patient from first contact through structured intake, AI-assisted acuity scoring with plain-language and technical explainability, confidence-fusion between the model and the nurse's own certainty, continuous post-triage deterioration monitoring, dynamic queue and bed orchestration, and a fully auditable trail of every override and escalation. It is delivered as a single, dependency-free HTML file that runs in any modern browser, which made it possible to build and rigorously test a complete, realistic product experience within the scope of a challenge submission, without standing up any backend infrastructure.

155M
Emergency department visits in the United States in 2022
Source: CDC / NCHS Data Brief, 2022 ED visit rates

90%+
Of U.S. emergency departments routinely report crowded conditions
Source: American College of Emergency Physicians (ACEP)

~25%
Of ED visits were over-triaged in a 5M+ visit ESI accuracy study
Source: Kaiser Permanente Northern California, 2016–2020

This proposal makes the case for what comes next: a phased path from working prototype to a piloted, clinically validated, EHR-integrated product — and an honest accounting of the safety, regulatory, and adoption risks that a program like this must manage deliberately from day one.

SECTION 2

Problem Framing

2.1 The scale of the problem

Emergency departments are both the busiest and the most operationally fragile part of the hospital. In 2022, U.S. emergency departments recorded 155 million visits — roughly 47 visits for every 100 people in the country [1]. The American College of Emergency Physicians reports that more than 90% of emergency departments routinely report crowded conditions, driven primarily by *boarding*: holding admitted patients in the ED because no inpatient bed is available [2]. A 2024 study cited by ACEP found that boarding patients in the emergency department nearly doubles the daily cost of their care [2]. Crowding is not an occasional spike; for most EDs, it is the baseline operating condition.

Three distinct failure modes compound inside that crowded baseline, and each is a place where better tooling — not more headcount — can materially help.

2.2 Triage inconsistency

Triage is the single decision that determines how quickly every subsequent patient is seen, and it is made under time pressure, often within the first two to five minutes of contact, on incomplete information. A large accuracy study of the Emergency Severity Index (ESI) — more than five million patient visits at Kaiser Permanente Northern California between 2016 and 2020, reviewed by a panel of emergency physicians and nurses — found that roughly a quarter of visits were *over-triaged* (assigned higher acuity than warranted) and about 3% were *under-triaged* (assigned lower acuity than warranted) [3]. Over-triage wastes scarce physician-review capacity on lower-acuity patients; under-triage is the far more dangerous failure mode, because it means a genuinely sick patient waits in a lower-priority lane. The same study found that 98% of encounters fell into the middle three of the five ESI levels — exactly the band where acuity is hardest to separate from experience and gut feel alone, and where structured decision support has the most room to help.

2.3 Deterioration is not a one-time event

Standard triage happens once, at arrival. But a patient's condition does not freeze at that moment — it evolves while they wait for a bed, a physician, or test results, sometimes for hours. Early-warning-score research has consistently shown that clinical deterioration is frequently preceded by measurable, trackable changes in vital signs well before a rapid-response or code event is called, and that systems built to continuously reassess — rather than triage once and move on — catch more of those windows [4]. Most ED information systems, however, are built around the single triage event and do not give staff an easy, always-on view of who in the waiting room or a low-acuity bay is quietly trending in the wrong direction.

2.4 Operational blind spots

Even when clinical judgment is sound, a department without a shared, real-time picture of capacity — available beds by category, expected discharges, physician-review backlog, wait-time breaches against safety thresholds — cannot act on that judgment quickly. Queue and bed status frequently live in the charge nurse's head, a whiteboard, or a separate bed-management system that does not talk to triage. That gap turns a clinical delay into an operational one, and it is invisible to leadership until it shows up as a diversion event or a patient-safety report.

The common thread. All three failure modes share a root cause: decisions and their supporting evidence are scattered across a clinician's memory, a paper chart, and disconnected systems, instead of being captured, explained, and kept current in one place a whole care team can see. That is the specific gap PatientTriage.ai is designed to close — not by replacing clinical judgment, but by making the evidence behind it visible, continuously current, and easy to act on.

SECTION 3

Solution Design

3.1 What was built

PatientTriage.ai is a fully interactive, single-file prototype — no server, no database, no build step — that implements a realistic emergency department workflow across five workspaces: **Operations** (the hospital-wide command center), **Triage** (structured intake and acuity assessment), **Monitoring** (continuous post-triage deterioration detection), **Capacity** (bed and surge orchestration), and **Analytics** (human–AI agreement, override patterns, and audit governance). It was deliberately built to preserve and extend a working clinical logic core rather than to look like a chatbot wrapped around a hospital theme: every AI output appears as a structured recommendation, a risk score, an explanation, or an alert — never as a conversation.

The product follows one consistent pipeline for every patient, visible throughout the interface:

CAPTURE → **ASSESS** → **EXPLAIN** → **VALIDATE** → **MONITOR** → **ESCALATE** → **ORCHESTRATE** → **LEARN**

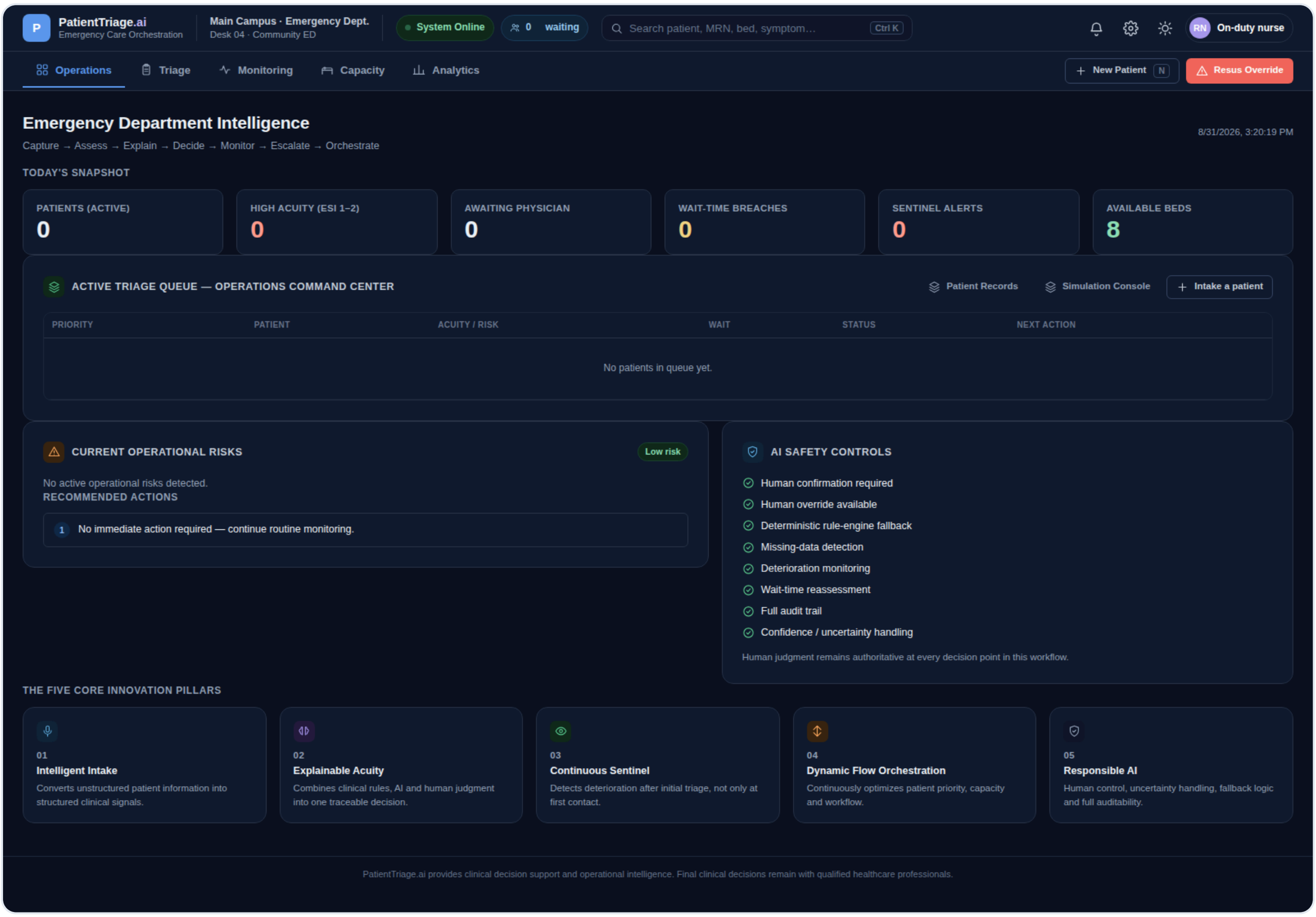

Fig. 1 — The Operations workspace: today's snapshot, active operational risks, recommended actions, and the command-center triage queue, all on one screen.

3.2 Five core innovation pillars

01

Intelligent Intake

Converts unstructured patient or nurse narrative into structured clinical signals, with a mandatory human-validation step before any AI-extracted field is accepted — nothing is auto-committed.

02

Explainable Acuity

Combines the deterministic clinical rule engine, the AI model, and the nurse's own assessment into one traceable recommendation, with a plain-language "why" and an optional technical (SHAP-style) breakdown.

03

Continuous Sentinel

Extends monitoring past the initial triage moment, tracking each waiting or boarded patient's deterioration risk trajectory and surfacing alerts before a rapid-response event, not after.

04

Dynamic Flow Orchestration

A real-time command-center queue and capacity view that continuously reflects acuity, wait-time safety thresholds, bed availability, and surge conditions in one place.

05

Responsible AI

Human confirmation is required at every decision point; overrides and escalations are always available and always logged; the system degrades gracefully to deterministic rules when AI is unavailable.

3.3 Decision support, not decision replacement

The centerpiece of the Triage workspace is a decision-support panel that separates **provenance** — what the clinical rules concluded, what the AI model concluded, and what the nurse concluded — before showing a single final decision. Confidence is treated as a first-class signal rather than hidden inside a black box: the interface visually fuses the AI's stated confidence against the nurse's own self-rated certainty (high or low/unsure) and turns that combination into a plain recommendation — proceed, seek physician review, or use clinical override — with the standing message that **human judgment remains authoritative**. When the AI service is unreachable or not configured, the interface does not pretend otherwise: it shows an explicit "AI service unavailable" state and confirms, in the acuity source itself, that the recommendation now comes from the deterministic rule engine or an offline heuristic fallback — verified end-to-end in automated testing of this prototype.

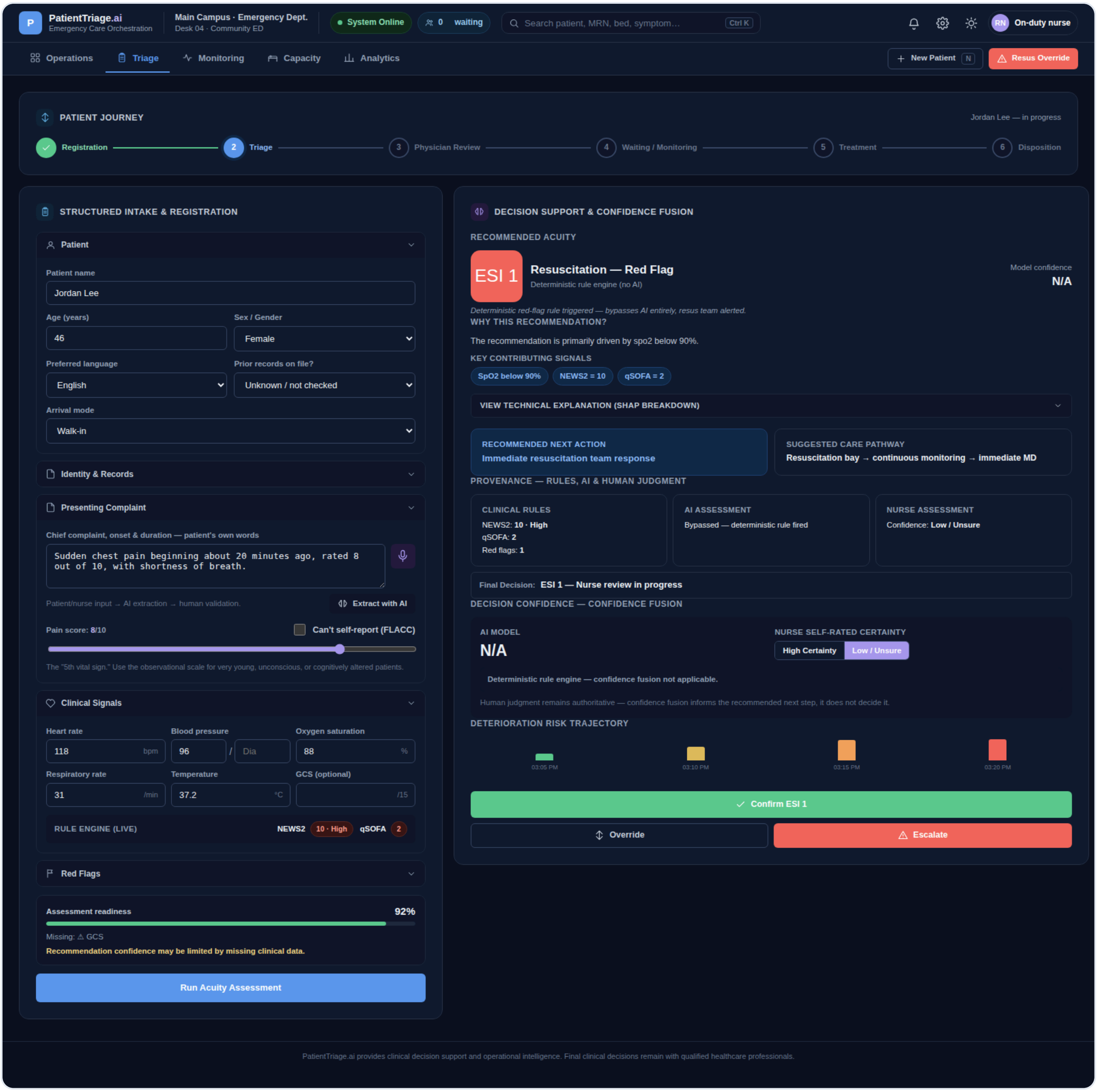

Fig. 2 — Decision Support & Confidence Fusion: recommended acuity, contributing signals, provenance across rules/AI/nurse, and the confirm / override / escalate action row.

3.4 From triage moment to continuous care

Once a patient is triaged, the Continuous Deterioration Monitor keeps tracking them — vitals, ESI, a rolling risk trajectory, and trend direction — and raises an alert with an acknowledge action when a patient trends toward higher risk, closing the gap described in Section 2.3. Every priority change carries a "why did this change" timeline entry, so a physician glancing at a patient mid-shift can see not just the current state but the reasoning trail that produced it.

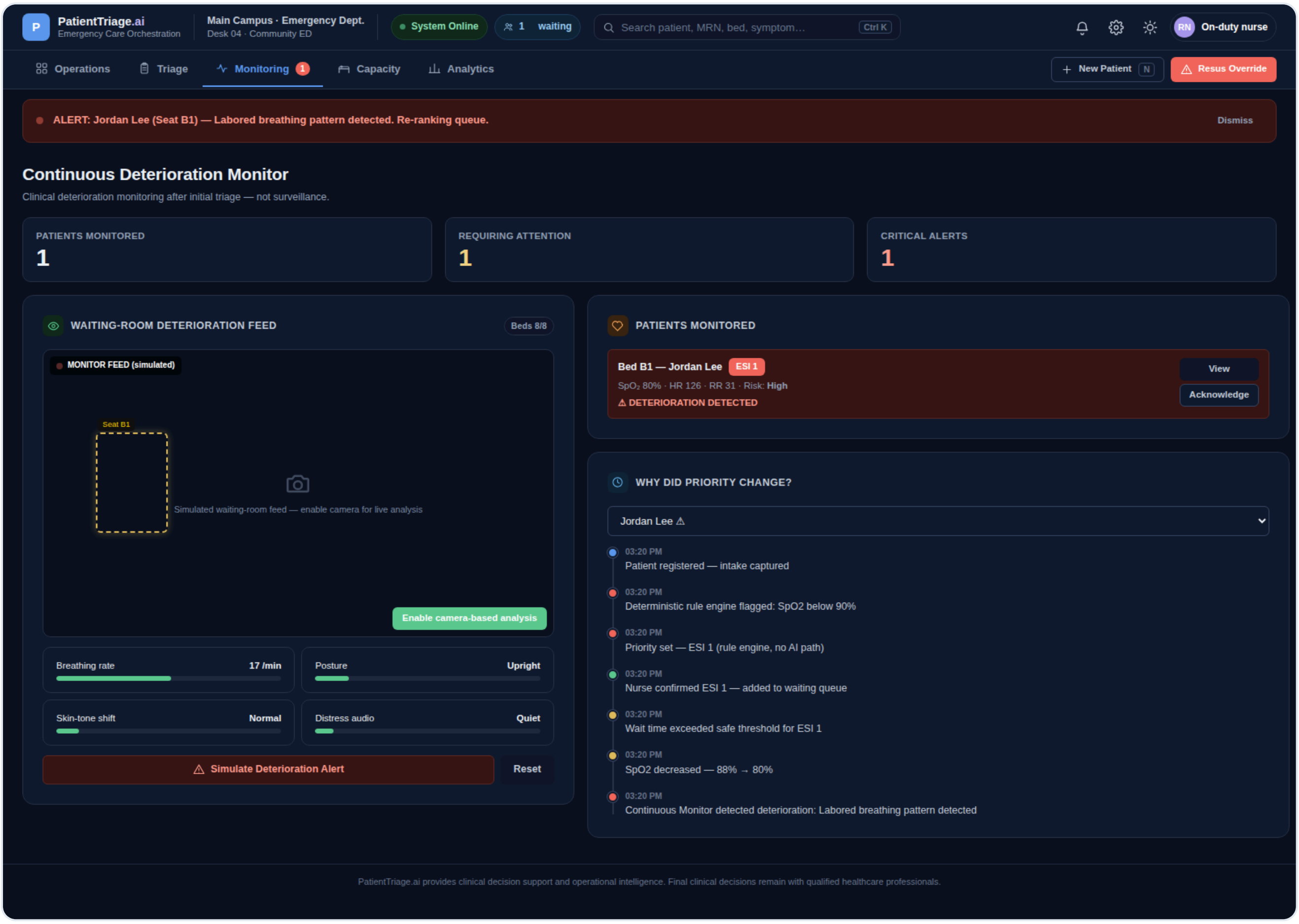

Fig. 3 — Continuous Deterioration Monitor: patients-monitored / attention / critical counts, and a per-patient monitored list with trend and alert state.

3.5 Built and tested like a product, not a mockup

The prototype was validated with an automated, browser-driven QA pass covering the full patient journey, a high-acuity and a normal-acuity patient, a manual override, a manual escalation, a simulated deterioration alert, wait-time breach handling, bed assignment, surge-mode activation, the AI-unavailable fallback path, dark and light themes, and tablet and mobile responsive layouts — with zero unresolved console or page errors at the end of that pass. The QA scripts themselves ship in the source repository so this can be independently re-verified.

SECTION 4

Target Users

The product is designed around the people who actually work an ED shift, plus the leaders who are accountable for how that shift performs as a system.

4.1 Primary users

Triage Nurse

PRIMARY USER · TRIAGE WORKSPACE

Owns the first clinical contact with every patient. Needs to move fast without skipping a safety check, wants a second opinion she can see the reasoning behind, and needs an override path that takes seconds, not a phone call, when her own judgment disagrees with the system.

Emergency Physician

PRIMARY USER · MD GLANCE VIEW, MONITORING

Needs a one-click, physician-formatted summary of a patient — complaint, vitals, NEWS2/qSOFA, AI assessment and key drivers, nurse assessment, current status — to triage his own review queue across many patients at once, and needs deterioration alerts that are trustworthy enough not to be ignored.

Charge Nurse / Bed Manager

PRIMARY USER · OPERATIONS, CAPACITY

Owns flow for the whole department during a shift — the queue, bed assignment, and surge response. Needs one real-time view of acuity, wait-time breaches, and available capacity instead of piecing it together from a whiteboard and three different screens.

4.2 Secondary users

ED / Hospital Operations & Quality Leadership

SECONDARY USER · ANALYTICS, AUDIT LOG

Accountable for throughput, safety events, and regulatory audit readiness. Needs override and escalation patterns, human–AI agreement trends, and a defensible, complete audit trail — not a system that quietly makes decisions no one can later explain.

Health System Executives

SECONDARY USER · BUSINESS CASE SPONSOR

Evaluating whether AI-enabled clinical tooling is a safe, defensible investment. Needs a system whose responsible-AI posture — human-in-the-loop by design, deterministic fallback, full auditability — can be explained plainly to a board, a regulator, or a plaintiff's attorney.

4.3 Tertiary: the delivery partner

For Accenture specifically, PatientTriage.ai is designed to also function as an accelerator: a pre-built, clinically-grounded reference implementation that a healthcare delivery team can bring into a hospital transformation engagement, rather than starting a clinical-AI conversation from a blank page. The responsible-AI posture built into the product — human confirmation required, deterministic fallback, full audit trail — is itself a reusable point of view that shortens the trust-building phase of an enterprise health system engagement.

SECTION 5

Business Case & Impact

5.1 Market context

The AI-enabled clinical decision support market was estimated at roughly $1.3 billion in 2025, projected to grow to approximately $4.5 billion by 2033 — a compound annual growth rate near 17% — driven by continued EHR adoption, the shift toward value-based care, and pressure to improve clinical workflow productivity [5]. Emergency and acute care is one of the highest-urgency segments of that market, precisely because the cost of a slow or wrong triage decision compounds quickly, and because EDs are already instrumented with vitals data that a decision-support layer can use immediately.

5.2 Where the value comes from

Four value drivers map directly to the problems in Section 2:

- **Reduced door-to-triage and door-to-physician-review time** for high-acuity patients, from structured, AI-assisted intake and a decision-support panel that leads with a next-best-action instead of a raw score the nurse has to interpret alone.
- **More consistent acuity assignment** across nurses of varying experience levels, addressing the roughly 25%/3% over-/under-triage pattern documented in Section 2.2, by giving every nurse the same rule engine and explainable AI baseline, with her own judgment layered on top rather than replaced.
- **Earlier detection of in-department deterioration**, closing the gap between the single triage event and the hours a patient may spend waiting, boarded, or in a low-acuity bay before their condition is reassessed.
- **Lower liability and audit-preparation cost** from a complete, structured record of every override, escalation, and its clinical rationale — work that is currently reconstructed manually from charting and staff recollection after an adverse event.

5.3 Illustrative impact model

The figures below are **illustrative** planning assumptions for a mid-sized community ED (150–300 visits/day, matching the prototype's default facility profile), intended to frame the size of the opportunity for a pilot business case — not measured outcomes, and not a guarantee. They should be replaced with real figures from a shadow-mode pilot (Phase 1, Section 6) before being used in any external or regulatory communication.

Metric	Illustrative baseline	Illustrative target after pilot	Basis
High-acuity patients receiving a decision-support recommendation within 5 minutes of arrival	Ad hoc	≥90%	Structured intake + AI-assisted scoring path built into the prototype

Metric	Illustrative baseline	Illustrative target after pilot	Basis
Under-triage rate on ESI 1–2 equivalent presentations	~3% (published ESI study, Section 2.2)	Directionally lower, pending pilot measurement	Confidence-fusion escalation path; not a claimed clinical outcome
Deterioration alerts acknowledged before a rapid-response event	Not systematically tracked today	Newly measurable via Monitoring workspace	Continuous Sentinel risk trajectory
Time to produce an audit-ready override/escalation record	Manual chart review, hours	Minutes (structured export)	Built-in Audit Log

5.4 Why this is an Accenture-shaped opportunity

The hardest parts of taking a prototype like this into production are not visible in a demo: HL7/FHIR integration with incumbent EHRs, clinical validation studies, change management across nursing and physician staff who are (rightly) skeptical of new triage tooling, and navigating the regulatory question of whether a given feature set functions as clinical decision support or as a regulated medical device. These are exactly the systems-integration, clinical-transformation, and regulatory-navigation capabilities a large consulting and technology partner brings, and they are the reason this proposal frames the prototype as a starting accelerator rather than a finished product.

SECTION 6

Phased Roadmap

The roadmap below is intentionally conservative about clinical claims: nothing beyond Phase 0 has been validated in a real clinical environment, and Phase 2 exists specifically to produce that validation before any wider rollout.

Phase	Timeframe	Goal	Key milestones
Phase 0 Prototype Complete	—	Prove the concept end-to-end	Working, fully-tested single-file prototype; five-workspace IA; responsible-AI safety controls designed in from the start; complete automated QA pass
Phase 1 Shadow-mode pilot	Months 0–6	Validate against real (de-identified) clinical scenarios without affecting live care	Single-site partnership; recommendations generated and logged in parallel with existing triage, never acted on directly; clinician feedback loop; formal safety and bias review; legal/regulatory classification assessment
Phase 2 Integration & clinical validation	Months 6–14	Connect to real systems and prove clinical accuracy	HL7/FHIR integration with the pilot site's EHR and vitals monitors; formal clinical validation study against physician-adjudicated outcomes; refined confidence-fusion thresholds using real agreement data; data privacy/security architecture and BAA execution
Phase 3 Guided live rollout	Months 14–24	Move from shadow mode to advisory use in live care at the pilot site, then expand	Regulatory pathway decision (CDS exemption vs. SaMD clearance) finalized with legal counsel; live advisory rollout with human-in-the-loop guardrails; second and third site onboarding; benchmarking dashboard for override/agreement analytics across sites
Phase 4 Scale & continuous learning	Months 24+	Multi-site platform with a durable feedback loop	Expansion to additional facilities and facility types (urgent care, inpatient deterioration monitoring); structured model-improvement loop from override/escalation data; packaging as a repeatable Accenture health-system offering

Design principle carried through every phase. The human-in-the-loop and deterministic-fallback architecture built into the Phase 0 prototype is not a stopgap to be removed later — it is the permanent safety backbone of the product, and every later phase is required to preserve it.

SECTION 7

Key Risks & Mitigations

Risk	Severity	Mitigation
Patient safety impact from an incorrect AI recommendation	High	Human confirmation is required before any acuity is assigned; override and escalation are available at every decision point; the deterministic clinical rule engine (ESI/NEWS2/qSOFA logic) provides an always-available, non-AI baseline that the AI recommendation is checked against, not a replacement for it.
Alert fatigue from the Continuous Deterioration Monitor	High	Tiered alerting (attention vs. critical) rather than a single undifferentiated alarm; confidence and trend-based thresholds tuned during the Phase 1 shadow-mode pilot using real clinician feedback before any alert can affect live workflow.
Clinician distrust or workflow disruption	Medium	Shadow-mode pilot before any live use; every recommendation ships with a plain-language explanation and an optional technical breakdown; override takes seconds and is never penalized in the design — the product is built to earn trust, not assume it.
Data privacy and regulatory exposure (HIPAA and equivalent)	High	The prototype itself never transmits or stores real patient data; production architecture will require a formal security review, signed Business Associate Agreements, encryption in transit and at rest, and role-based access control before any real data is processed.
Uncertain regulatory classification (clinical decision support vs. regulated medical device)	Medium	Legal and regulatory counsel engaged early in Phase 1; product design deliberately keeps the human as the final decision-maker on every output, which is the central factor most CDS regulatory frameworks weigh; feature scope adjusted if a given capability would cross into device-level claims.
EHR and medical-device integration complexity	Medium	Phase 2 scopes integration explicitly, using standard HL7/FHIR interfaces and a single pilot site before any multi-site rollout, rather than assuming integration effort away.
Algorithmic bias across patient demographics	Medium	Clinical validation study in Phase 2 explicitly includes a fairness/bias review across age, sex, and other demographic groups before any live rollout; deterministic rule engine provides a bias-resistant floor that AI recommendations are measured against.
Dependency on a third-party AI model provider	Low	Already mitigated in the Phase 0 prototype: when the AI service is unavailable, misconfigured, or unreachable, the system automatically and visibly falls back to a deterministic heuristic scorer, verified in automated testing — the department is never left without decision support.

Risk	Severity	Mitigation
Over-reliance on AI eroding clinical skill over time	Medium	Product language and UI consistently frame outputs as "decision support," never "the answer"; nurse self-rated certainty is an explicit, required input to the confidence-fusion logic, keeping active clinical judgment in the loop rather than passive approval.

SECTION 8

Conclusion & Ask

Emergency departments do not need another dashboard, and they do not need an AI system that quietly makes decisions on their behalf. They need the evidence behind a triage decision to be visible, current, and easy to act on — from the first two minutes of contact through however many hours a patient waits for a bed. PatientTriage.ai demonstrates, as a working and tested prototype, that this is buildable today with a human-in-the-loop architecture that does not trade safety for speed.

The ask of this proposal is straightforward: treat the Phase 0 prototype as validated proof of concept, and fund a Phase 1 shadow-mode pilot with a single willing hospital partner. Shadow mode means real risk stays at essentially zero — the system runs alongside existing triage, not in place of it — while producing the first real evidence needed to responsibly move toward Phase 2 clinical validation and integration.

The complete source code, the automated QA suite that validated it, and this document are provided together so the underlying engineering, not just the pitch, can be evaluated on its merits.

PatientTriage.ai is a demonstration prototype. It is not a certified medical device and has not been clinically validated. All in-product performance figures not attributed to a cited external source are illustrative simulations, clearly labeled as such within the product itself, and must not be treated as measured clinical outcomes.

SECTION 9

Sources

1. Centers for Disease Control and Prevention, National Center for Health Statistics. *Emergency Department Visit Rates by Selected Characteristics: United States, 2022.* NCHS Data Brief. 155 million ED visits recorded in 2022; overall visit rate of 47 per 100 people. ncbi.nlm.nih.gov/books/NBK607412
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4. Systematic review literature on early warning scores and failure-to-rescue in hospitalized and emergency department patients, summarized in: *Early Warning System Scores for Clinical Deterioration in Hospitalized Patients: A Systematic Review*, Annals of the American Thoracic Society; and *The use of early warning system scores in prehospital and emergency department settings to predict clinical deterioration: A systematic review and meta-analysis*, PMC. pmc.ncbi.nlm.nih.gov/articles/PMC8929648
5. Grand View Research. *Artificial Intelligence in Clinical Decision Support Market Report.* Market estimated at USD 1.3 billion in 2025, projected to reach USD 4.5 billion by 2033 at a 17.1% CAGR (2026–2033); North America held 55.9% revenue share in 2025. grandviewresearch.com/industry-analysis/artificial-intelligence-clinical-decision-support-market-report

Figures were current as of the search dates noted in the accompanying research (August–September 2026) and are subject to revision by their publishers. This proposal cites them as directional industry context, not as claims verified by the author.